

# T-Markov

SKT Global Intelligent Network & Security Platform

2025.03

SKT Enterprise

SKT Enterprise

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# 01

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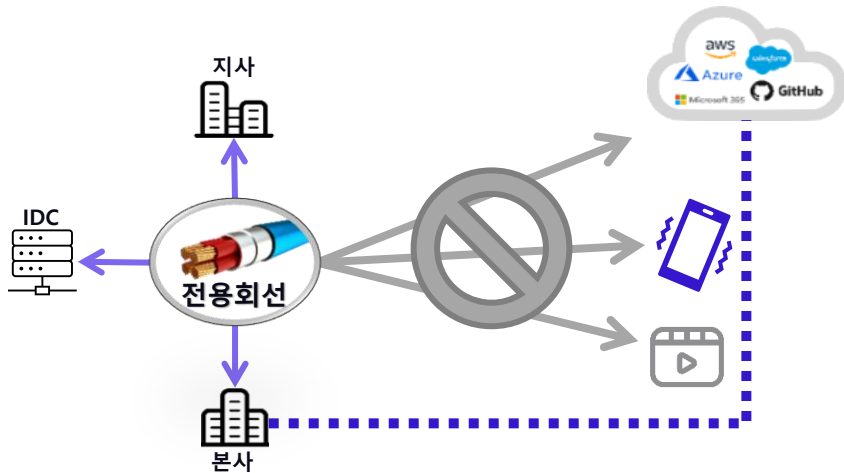
Background

# 01 Background

## Transformation : Virtualization of Network Connectivity

Increased demand for software-based, flexible connectivity and security from fixed equipment and connectivity

### Legacy Network/Security



Limitations of Legacy Network/Security

AI + Virtualization

+

Cloud/SaaS Connection

+

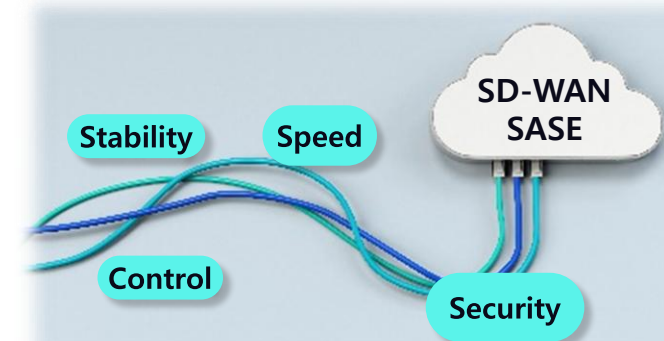
Simplify Complexity  
Cost saving

+

Respond immediately to  
security threats

Customer Needs

Virtualization of the network is not an option, but a necessity



"Virtualized Network/Security"  
The Emergence of Platform Services

# 01 Background

## Change : Intelligent, Diversified Security Threats

상반기 침해사고 최다 발생 업종...‘제조업’, 주로 ‘크리덴셜 스티핑’, ‘VPN’ 취약점 악용  
(한국인터넷진흥원) 2022. 08. 24

"Credential stuffing" breaches using VPN vulnerabilities spiked  
One vulnerability exposes all confidential information on the internal network



[긴급] KAI도 원자력연 당했던 VPN 취약점으로 해킹 당했다 <머니투데이> 2022-11-21

"Domestic government and corporate VPN services are not secure"  
KAI, Han Won-yeon, and other institutions with technical information are targeted



다.

Ransomware infections spike through Cisco VPN vulnerability  
Global Standard as a vulnerability...



# 01 Background

## The Era of SD-WAN & SASE

SD-WAN and SASE Market are expected to grow at a high CAGR of 33% and 26%, respectively, in '2024-'2031.

\* Source: verified market research

### SD-WAN

- Includes CSP, DC, and SASE manufacturers in addition to Telco
- Using SD-WAN equipment and Global PoP, we provide quick and easy connection of domestic and overseas dedicated lines

\* SD-WAN : Software Defined WAN

### SASEZTNA SDP

- Government-led push to make ZTNA mandatory
- The best solution to prevent continuous hacking incidents using VPN service vulnerabilities

- \* SASE : Secure Access Service Edge
- \* ZTNA : Zero Trust Network Access
- \* SDP : Software Defined Perimeter

From the existing method of providing network, security equipment, and line services individually, to providing all services with an integrated platform

SD-WAN & SASE are『 Global Mega Trend 』



# 01 Background

## SKT's 13-year+ journey



After the cold season of AI, the world's first AI-based network platform was implemented.  
Looking at the future from the boundaries of the present with memories of the past

# 01 Background

Changing the environment: AI-based, intelligent and diversified security risks



TCP packet problem prediction and Real-time network alignment  
Average 370% speed  
Standard deviation greater than 90% improvement



AI-Powered Global Network Configuration learning and real-time paths update  
100% SLA Compliance (2018~)



Special encrypted tunnel  
Seamless turnover.  
Breakthrough stability + security  
4x+ improvement in satellite network speed  
Real-time tunnel changes



With AI-powered XDR  
Zero Trust  
Complete active security system  
  
Over 3500 nodes  
Real-time surveillance

- Network Packet Deep Learning Analysis
- Automatic route switching and optimization
- Specialized encryption tunnel technology



- Implementing a Zero Trust Environment
- AI-Powered XDR Integration
- Real-time threat detection and response



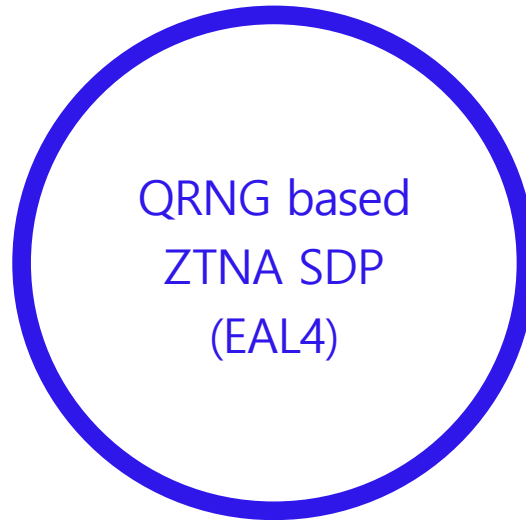
# 02

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## Overview

## 02 Overview

T-Markov is an intelligent network & security solution that improves network performance and provides security features at the same time.



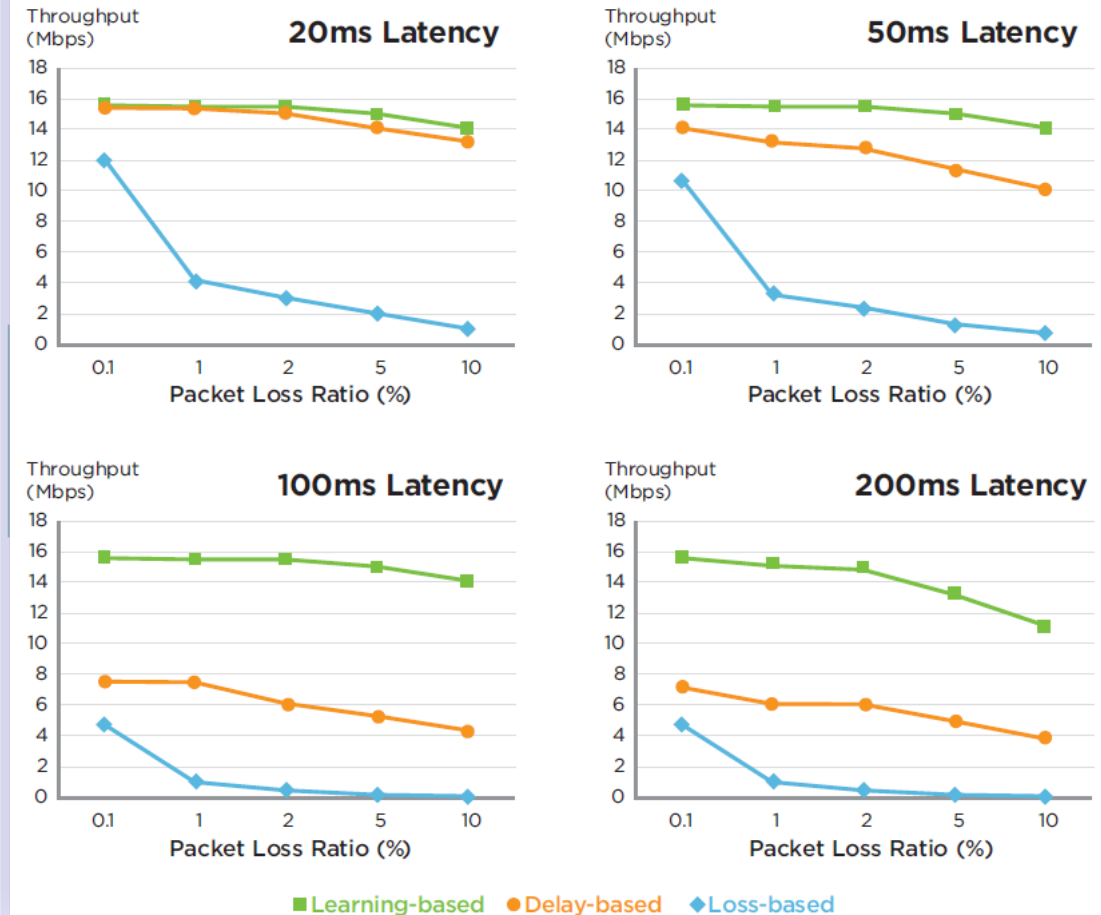
*"SK's Readiness"*

## 02 Overview

### Learning-based TCP acceleration Technology

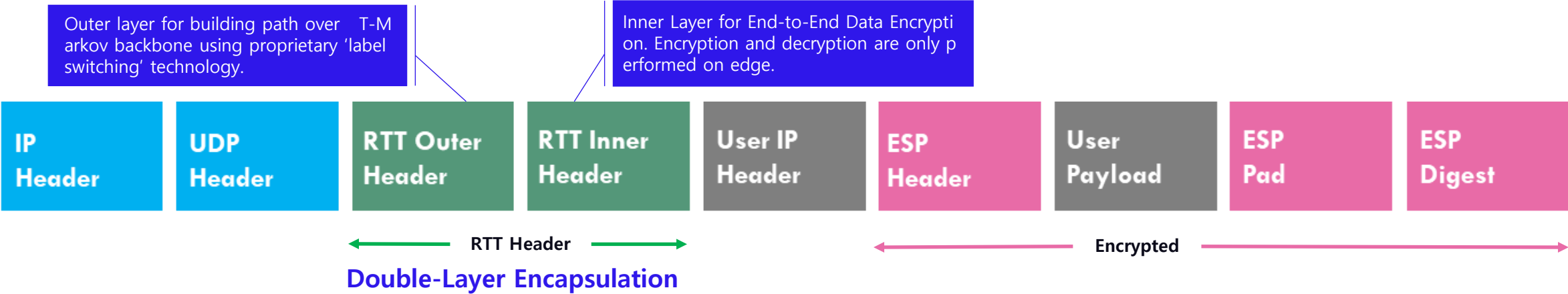
- ✓ **ZetaTCP™** is the leading learning-based TCP acceleration technology, widely used and/or licensed by large carriers worldwide.
- ✓ T-Markov **RTT<sup>Real-Time TCP</sup> tunnel** Integrates ZetaTCP™ algorithms and **significantly improves** transmission speed and stability of T-Markov.
- ✓ RTT is further enhanced with algorithms that enable **seamless path switch** without affecting applications/user experiences.
- ✓ Additionally, RTT includes **end-to-end encryption to ensure user data remains secure** during transmission over backbone network. With End-to-End Encryption, data is encrypted and decrypted solely on the CPE device which is deployed at customer's site, and the backbone PoPs will only forward the encrypted data from one point to another.

#### Third Party Comparison Tests



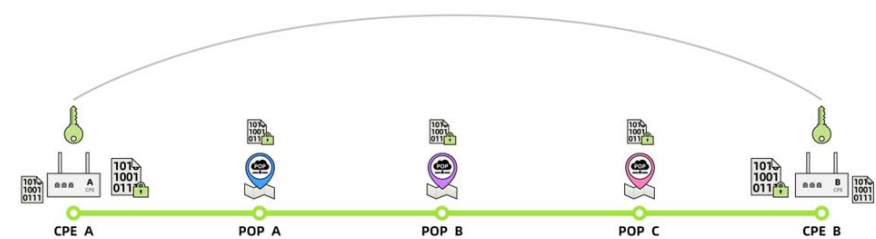
# 02 Overview

RTT Real-Time TCP - Security & Performance on a single stack



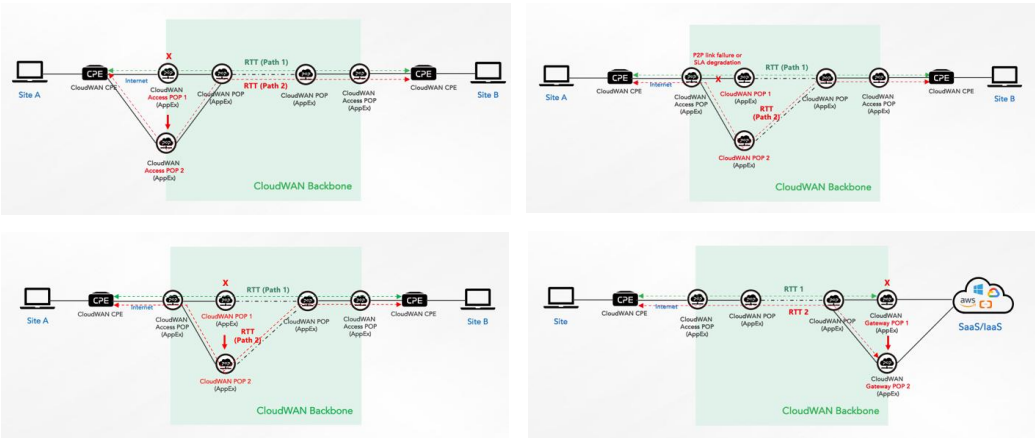
## Transmission Security

End-to-End Encrypted Tunnel



## Graceful Path Switchover

The Orchestrator keeps monitoring status/performance of tunnels and related resources and perform switchover when SLA is not met. The switchover is application agnostic.



# 02 Overview

T-Markov is the most complete enterprise secure networking platform

Reference

Numerous Global References



Compatibility

Compatible with all types of lines

CPE

More than 100 species  
Wired & Wireless Integrated CPE  
(Wired and Wireless, 4/5G)

Mobile CPEvCPE

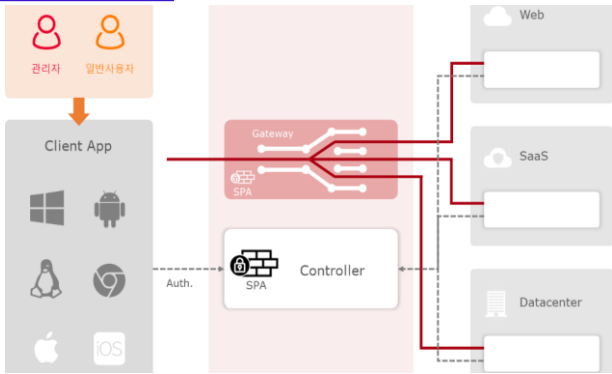
Personal/Business OS  
All supported  
Integrated Agent

vCPE

Commercial Cloud and  
All Compatible  
Virtual CPE

Security

Black Cloud based ZTNA



Manageability

The most advanced ZTP (Zero Touch Provision)



Enter CPE Activation URL,  
Set up in 1 second!!

Quality

The Most Mature AI Quality/Path Management



Quantum Security

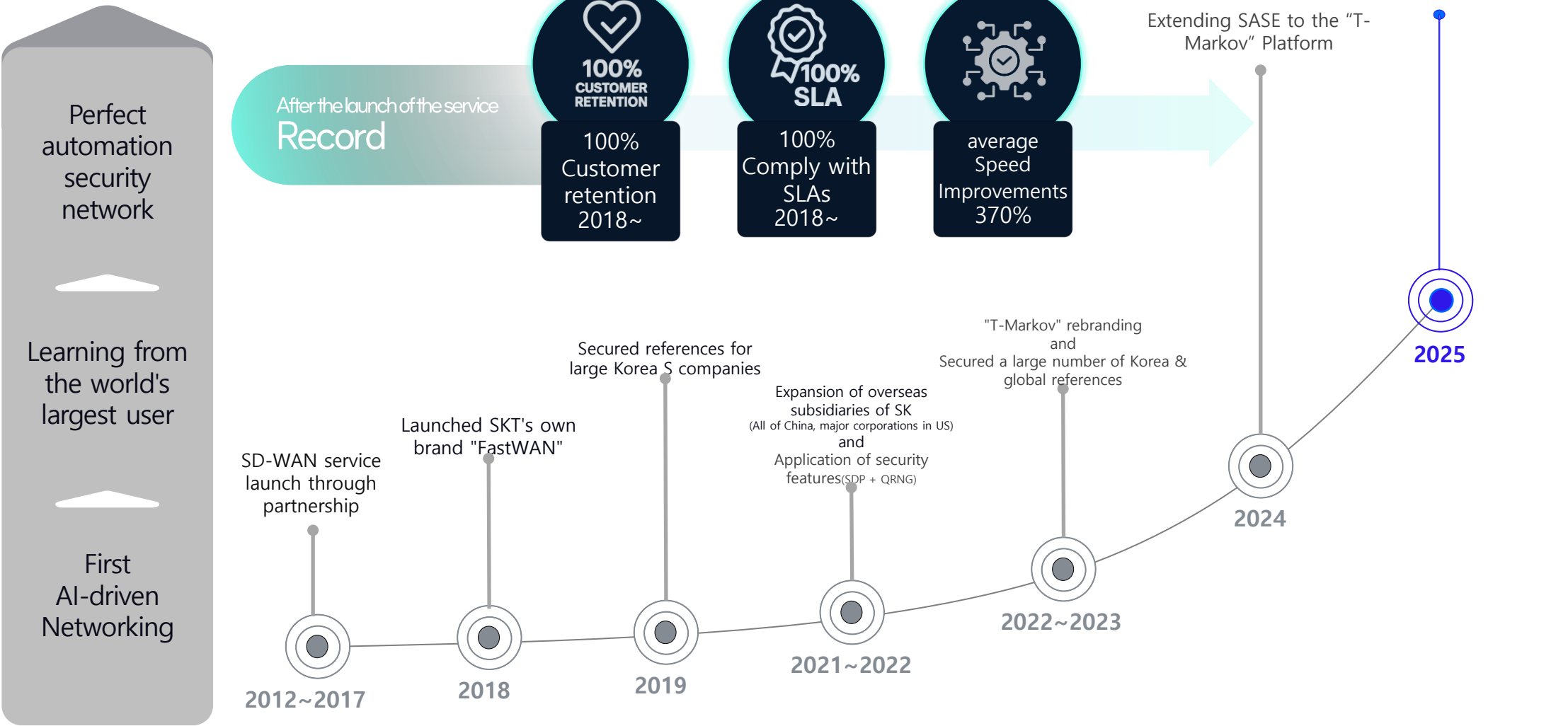
Enforce secure quantum cryptography keys





# 02 Overview

## T-Markov History



# 03

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Highlights

## 03 Highlights



### Easy connection

Connect instantly to any environment

### Widest network coverage

Top-level coverage with no shaded area

### Perfect security

Korea's first ZTNA CC certification and quantum encryption QRNG

### Innovative network performance

Eliminate existing network bottlenecks, dramatically improve speed

### Network Integration

Integration of heterogeneous network environments

### TCO Improvement

Lower TCO through network/security efficiencies

## 03 Highlights

Easy Connection : Connect instantly to any environment

### Every Wire/Wireless Devices



### Every Cloud & SaaS

All major Cloud and SaaS integrations  
Providing SLA-based services



### Real-Time Transparency



### Applicable to all enterprise networks



- Integrate with equipment platform from all vendors
- Seamless connection of all types of lines
- Compatible with L2~L7 switches

### Improved access quality to M365



- Secure access to Hop inside the Cloud
- Connecting Global SaaS through DNS Delegation
- Closed network and customizing

### Powerful Data View in AI-Driven Portals

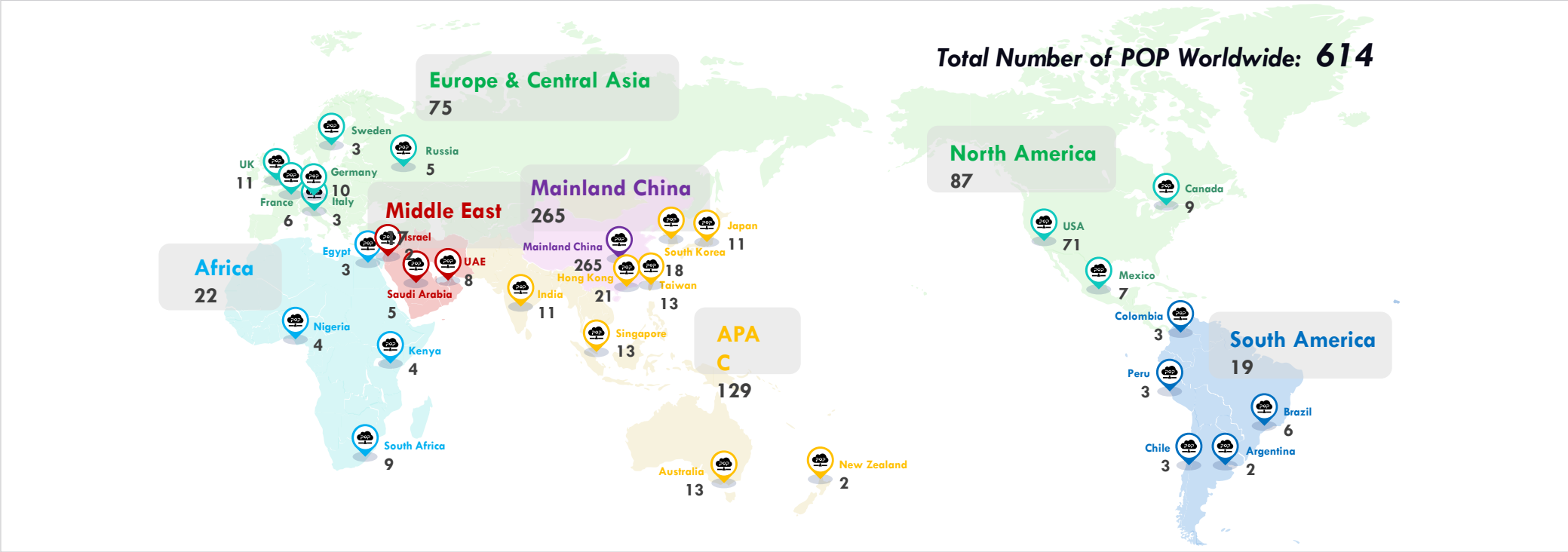


- All settings within the portal
- View all your data in less than 5 seconds
- Powerful operation conveniences

### 03 Highlights

Widest network coverage : More than 90% of the worldwide with low-latency access of less than 10ms

T-Markov Global Backbone - 600+ POPs Worldwide as of now



Real Global Coverage

Less than 10ms to 90% of population

AI Global Map

Expand with easy

New POP set up on the same day

Realtime Expansion

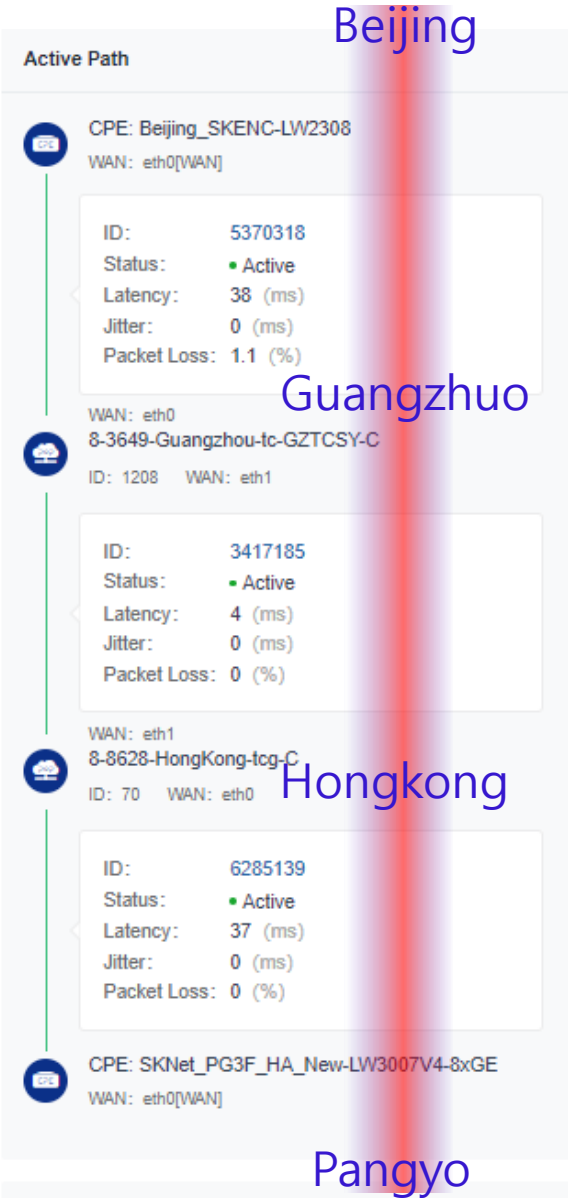
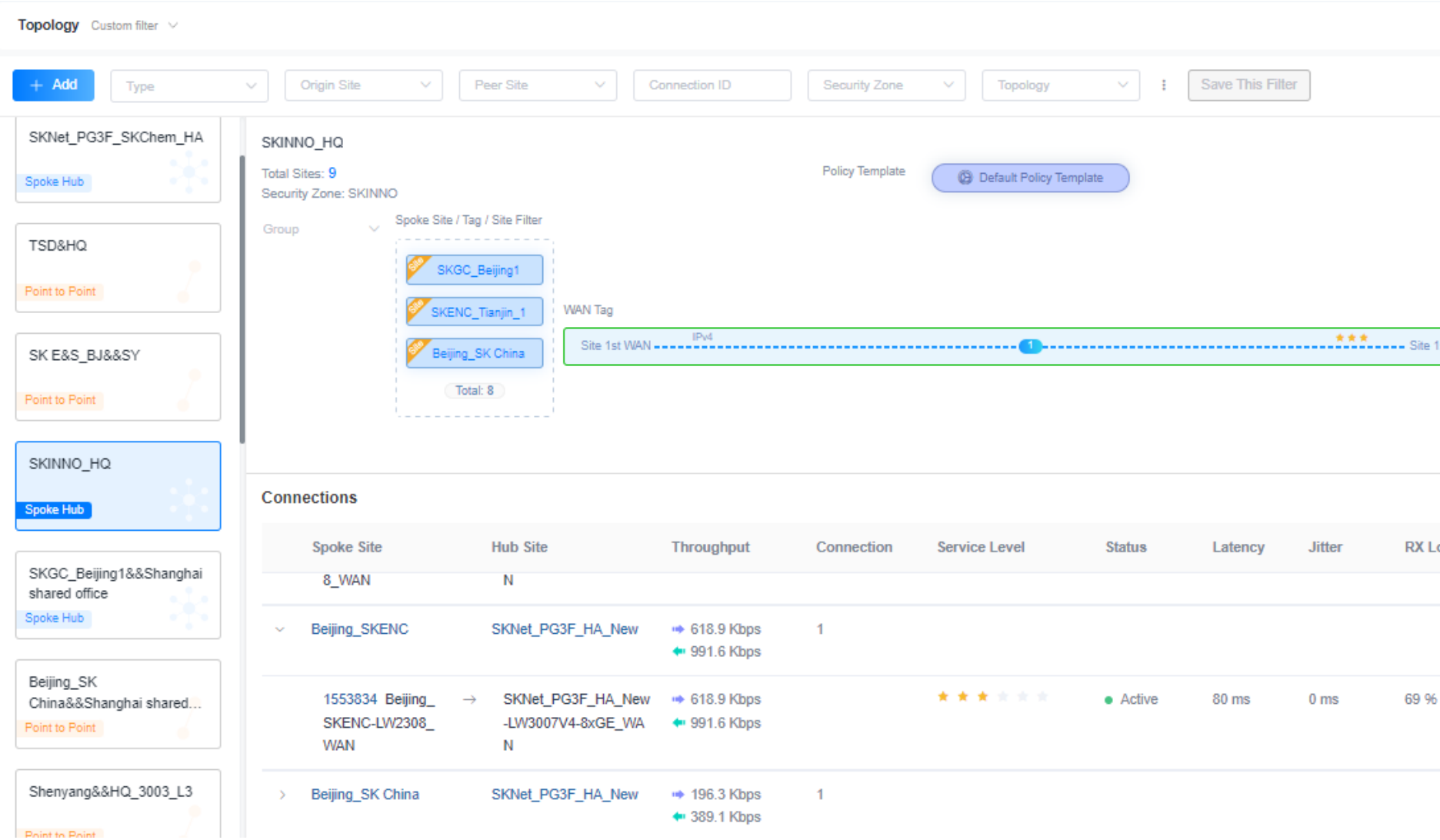
The only solution for

underdeveloped areas Africa, China, South America, etc.

High Coverage

# 03 Highlights

## Perfect security : Real-time Transparency & Visibility



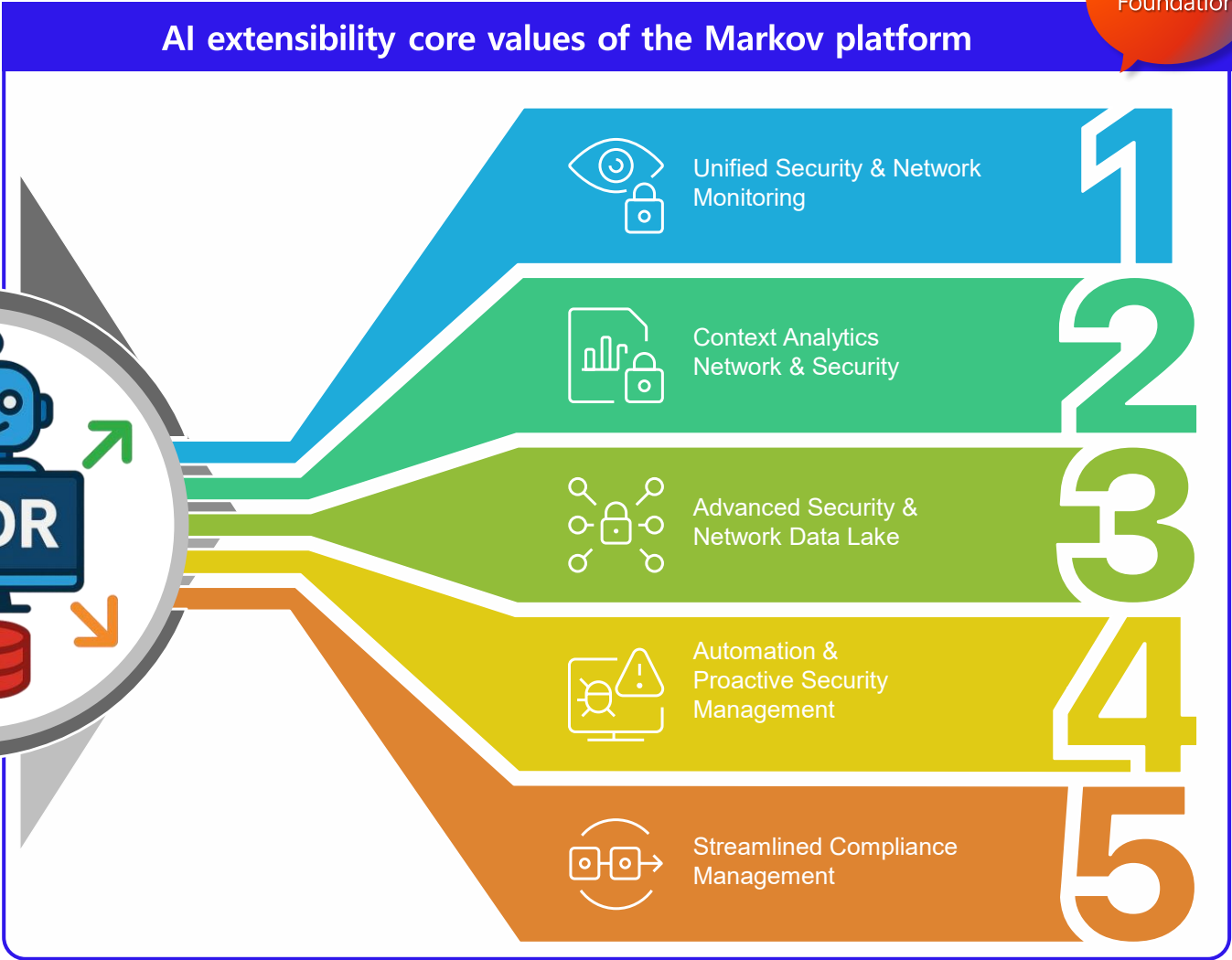
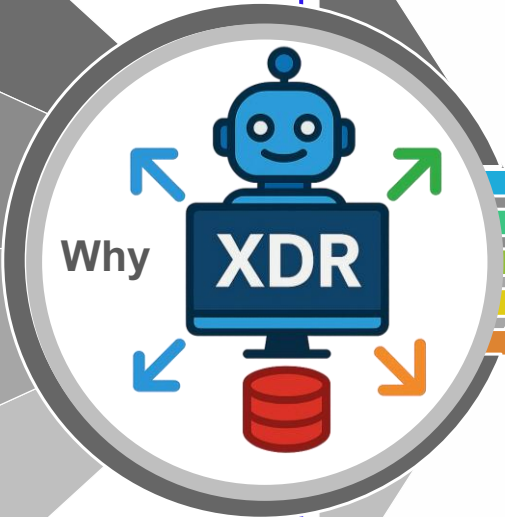
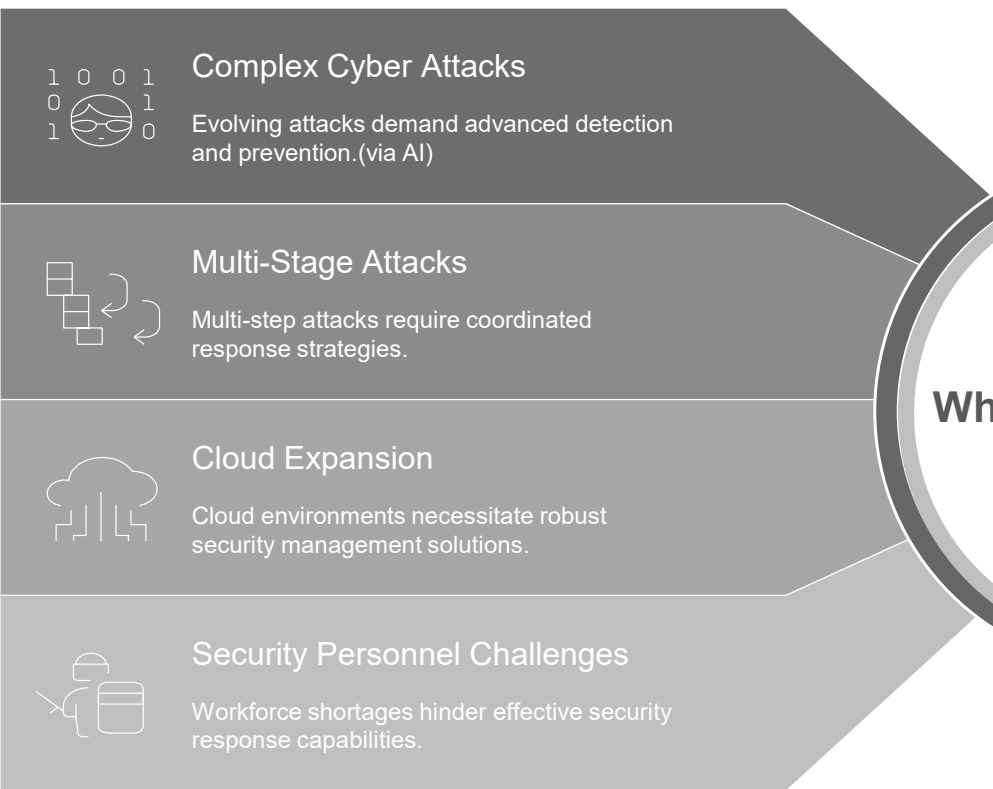


# 03 Highlights

“Addressing the Growing Complexity of Cybersecurity”

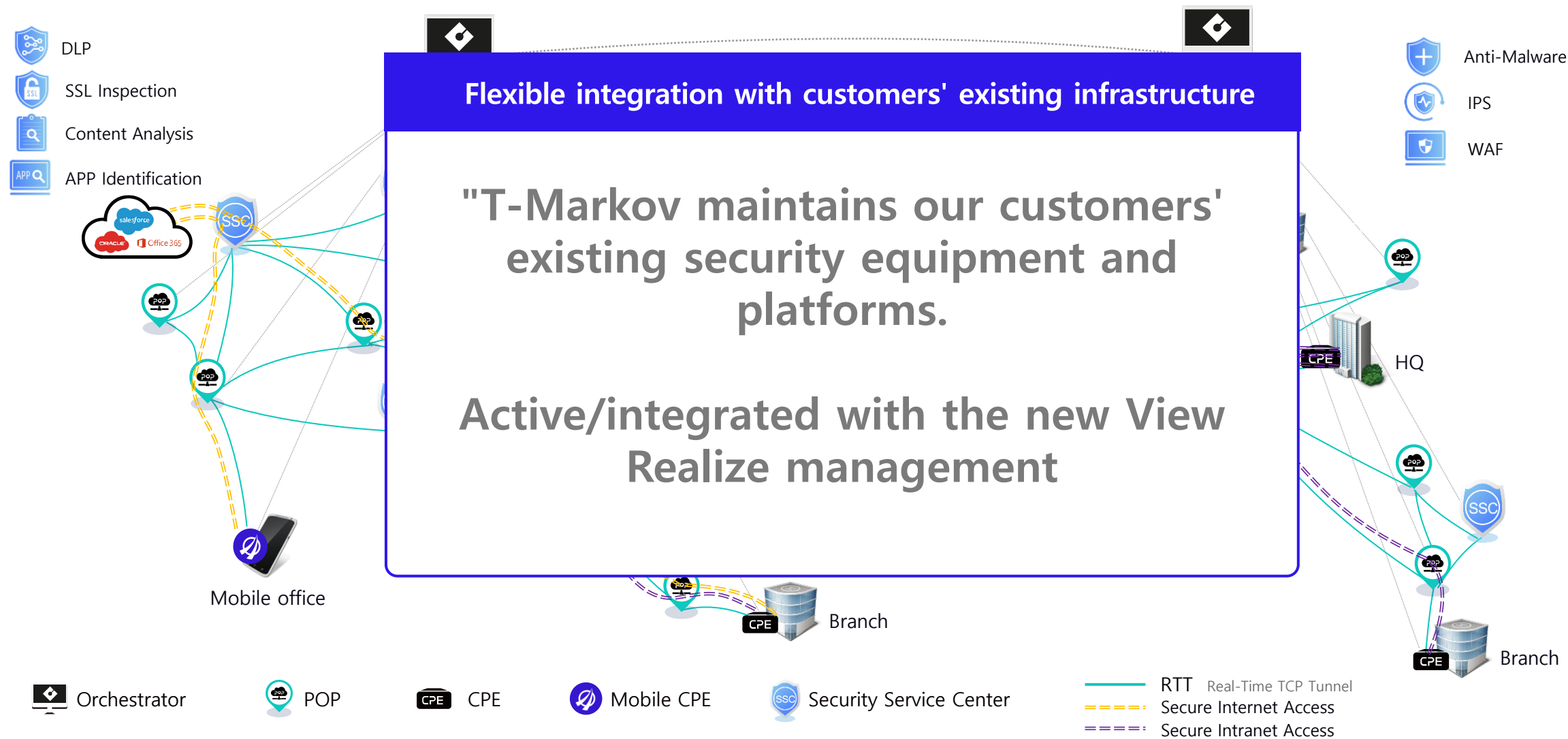
AI-Ready  
Foundation

## Evolution of Threat



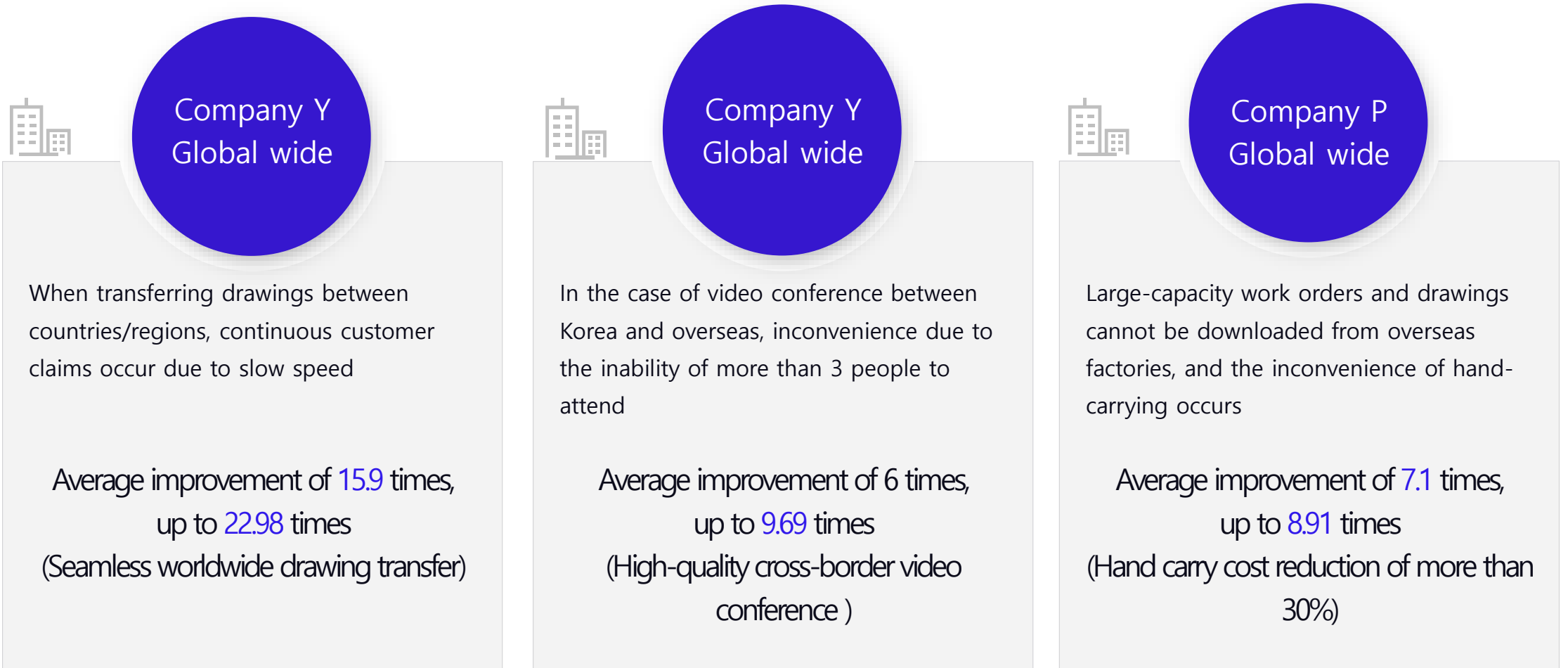
# 03 Highlights

## Implementing Active Security: Open SASE for Enhanced Security



### 03 Highlights

Innovative network performance : Eliminate existing network bottlenecks, dramatically improve speed



### 03 Highlights

Network Integration : Integration of heterogeneous network environments

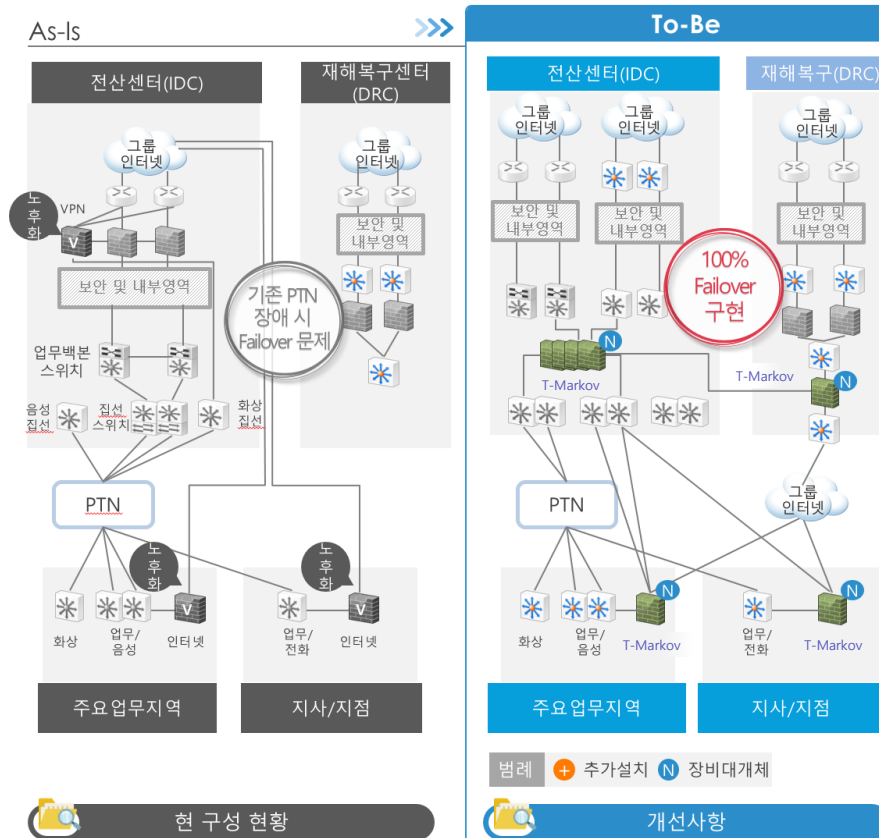
#### T-Markov Configuration



Company K  
Nationwide

Inconvenience of integrated management  
of the network of more than 120 business  
sites  
After network separation, there is a  
problem of distribution of leased lines and  
Internet traffic.

**Heterogeneous network integration**  
**100% failover in case of failure**  
**Applying policies for each site**



## 03 Highlights

TCO Improvement : Lower TCO through network/security efficiencies

### Network costs are increasing due to changes in the corporate environment

- Increasing network costs due to changes in the ICT environment such as **AI DC and Digital Transformation**
- Incurred **overhead costs** due to intrusion/infection caused by **vulnerabilities** in existing networks such as VPNs
- Increased **operating costs** due to the use of multi-type networks and **investment costs** due to network integration
- Increased **management costs** due to separate network and security management and the use of separate platforms
- Existing network speed limits lead to **work inefficiencies**

### Improving TCO with T-Markov

- **Lower TCO** with cost-effective T-Markov
- **Reduced overhead costs** due to increased security
- **Reduced operation/investment costs** due to network consolidation
- **Increase management efficiency** by using a integrated network and security platform
- **Increased work efficiency** by improving network speed

# 04

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Competitiveness  
(NDA)



# 05

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References

# 05 Reference (Overview)

T-Markov is used by more than 1,500 customers and more than 17% of Fortune companies worldwide.  
We secured numerous references in Korea including SK affiliates, and have completed verification of performance/security/stability.

Global & Korea











SKT Enterprise

## 05 Reference

Due to the aging of existing network/security equipment at SK Telecom's nationwide agencies, there is a need for 1) stable equipment replacement, 2) enhanced security, and 3) cost reduction. Excellent reference of SKT-SKB-SKC&C cooperation that solved network and security issues through SD-WAN and ZTNA based on SK own technology to agencies nationwide.

**Reliable replacement of existing equipment**

**Enhanced security for agencies(VPN->SASE)**

**Real-time monitoring/ response**

**Cost Savings(about 20%)**

01 진보한 SASE Platform 적용



02 ZTNA+(SDP) Platform 적용



03 보라매 대덕 Client DR 구성



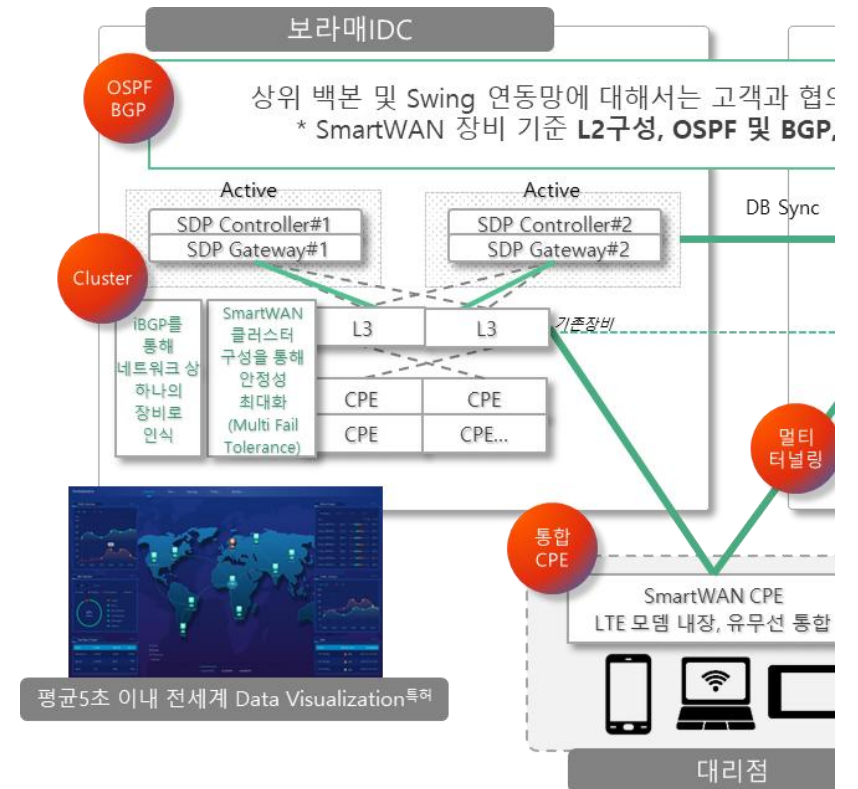
04 딥러닝 기반 Optimization



05 QoS & 사용자 컨트롤(STuple 기반)



06 Full Customizable 프레임워크



# 05 Reference

## Eliminate existing network bottlenecks, dramatically improve speed

Company A

Foundry



Slower drawing transfer speed compared to competitors  
Continuous occurrence of customer claims

- Available for: 700 GB drawings
- Method: Standard Download Tool
- Condition: 100 times average
- Compared to the existing dedicated line

### Worldwide Drawing Transfers

Company Y

Global wide



"Video conferences with Vietnam and Bangladesh are not allowed to have more than 3 people""Mr. Chairman, we held a meeting of factory managers around the world only with audio without video."

- For: Entire in-house system
- Method: Measure directly with employee browser tools
- Condition: More than 10,000 times in 2 months
- Compared to the existing leased line riverbed

### High-quality video conferences are underway with the participation of factory managers around the world

Company P

Global wide



Large work orders and drawings are difficult to download from the factory in Myanmar, Vietnam. I'm carrying a USB every day in person

- Available for: 1G, 10G, 50G, 100G Design Drawings
- Method: Pacific C&T Standard Download Measurement Tool
- Condition: 500 average calculations
- Conventional: KT Dedicated Line, Riverbed

### Reduced hand-carry costs by more than 30% and improved speed by 600%

| area               | San Antonio | California | Beijing | Faraday Shanghai | IBM Vermont |
|--------------------|-------------|------------|---------|------------------|-------------|
| improvement effect | X 8.85      | X 22.98    | X 22.25 | X 8.3            | X 17.5      |
| Max Improve        | x22.98      |            | Average | x15.9            |             |

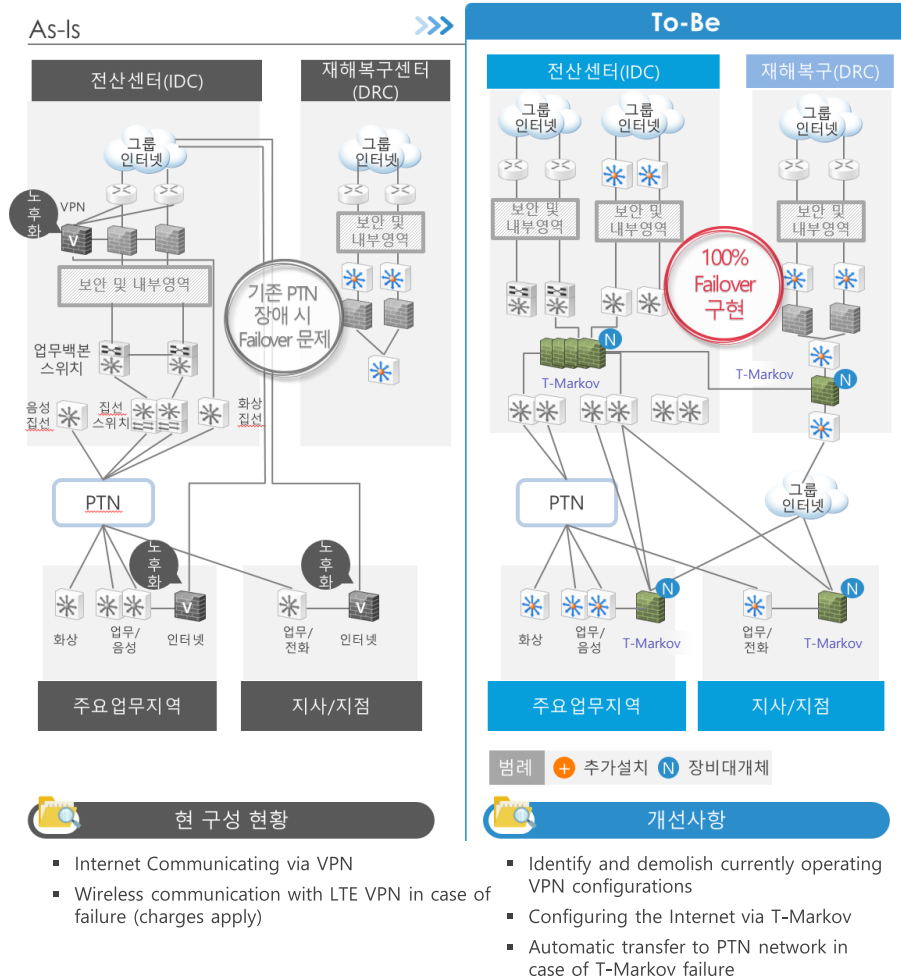
| area               | Vietnam            |      | Bangladesh         |      |
|--------------------|--------------------|------|--------------------|------|
| improvement effect | Average Throughput | 475% | Average Throughput | 682% |
| Max Improve        | 869%               |      | Average            | 502% |

| area               | Myanmar            |      | Vietnam            |      |
|--------------------|--------------------|------|--------------------|------|
| improvement effect | Average Throughput | 732% | Average Throughput | 577% |
| Max Improve        | 791%               |      | Average            | 613% |

## 05 Reference

Integrated leased line/Internet, switch/router/L4/VPN/firewall, etc., all into one network

### T-Markov Config



## Company K

### Customer Requirements

- Problems with the distribution of dedicated lines and Internet traffic after network separation
- Integrated management of more than 120 business sites nationwide
- 100% Failover in the event of a problem ...

### Customer Benefits

- T-Markov acts as the core backbone router
- Unified management of all dedicated/Internet lines
- 5Tuple-based QoS to apply policies for each business site



## 05 Reference

Improved speed and services access between Korea and China

### Korea S Bank : China Case Study

"Even though it is a leased line, it keeps cutting off and the quality is poor."

"It's very inconvenient to block certain sites."

"It's like auditing and blocking data."

"The delay M365 is too bad while using ."

Improved speed by an average of more than 20%

Average 7Mbps



The stuttering is gone, and I can work seamlessly in Word or PowerPoint."  
"It's really convenient to be able to Google and access the web hard of the headquarters."



# 06

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Why SKT?

## 06 Why SKT?

### ① Superior Network Technology

- Secured 6 U.S. patents related to AI acceleration and path optimization
- A number of contracts such as the Presidential Security Service based on superior technology

### ② Innovative Security Technology

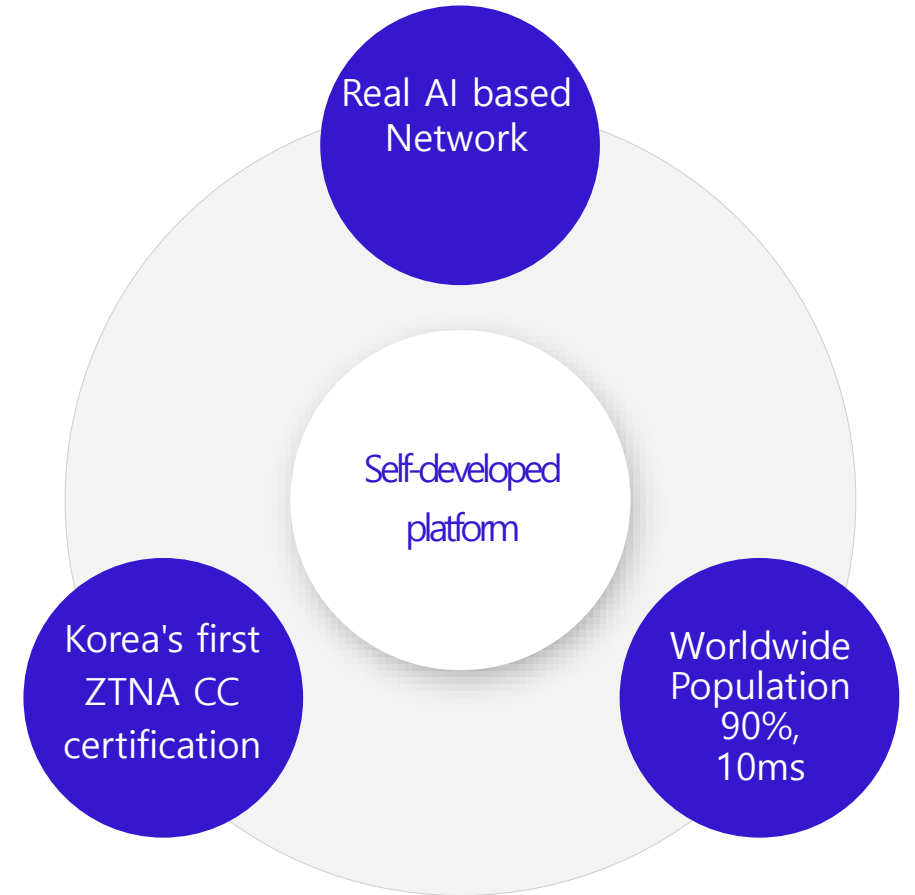
- The most impenetrable SDP method of ZTNA (Korea's first ZTNA CC certification)
- Readiness for Quantum cryptography QRNG

### ③ Global Top-level Network Coverage

- Great Firewall of China Risk Free (Acquired Tier1 business in China)
- Specialized in remote areas, overseas business trips, wireless environment

### ④ Self-developed platform and Business Scalability

- Many examples of on-premise implementation
- Full-customizable configurations to meet service requirements



Thank you.